



SHORE

Examination Number:

Set:

Year 12

HSC Assessment Task 5 - Trial HSC

17th August 2017

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen
- Board-approved calculators may be used
- A NESA reference sheet is provided
- Answer Questions 1–10 on the Multiple Choice Answer Sheet provided
- Start each of Questions 11–14 in a new writing booklet
- In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.
- Write your examination number on the front cover of each booklet
- If you do not attempt a question, submit a blank booklet marked your examination number and “N/A” on the front cover

Total marks – 70

Section I

Pages 3–6

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II

Pages 7–12

60 marks

- Attempt Questions 11–14
- Allow about 1 hour 45 minutes for this section

Note: Any time you have remaining should be spent revising your answers.

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Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the Multiple Choice Answer Sheet for Questions 1–10.

- 1 A function is represented by the parametric equations
 $x = 2t + 1$ and $y = t - 2$.
Which of the following is the Cartesian equation for the function?
- (A) $x - 2y + 3 = 0$
- (B) $x - 2y - 3 = 0$
- (C) $x + 2y + 5 = 0$
- (D) $x - 2y - 5 = 0$
- 2 Given that $\sin \alpha = \frac{4}{5}$ and $\cos \beta = \frac{5}{13}$ and both α and β are acute,
what is the exact value of $\cos(\alpha - \beta)$?
- (A) $\frac{-33}{65}$
- (B) $\frac{27}{65}$
- (C) $\frac{56}{65}$
- (D) $\frac{63}{65}$
- 3 What is the size of the acute angle, to the nearest degree, between the lines
 $2x - 3y + 4 = 0$ and $x + 2y - 7 = 0$?
- (A) 7°
- (B) 19°
- (C) 41°
- (D) 60°

4 What is the Domain and Range of $y = 3 \cos^{-1} 2x$?

(A) Domain: $-\frac{1}{2} \leq x \leq \frac{1}{2}$ and Range: $0 \leq y \leq 3\pi$.

(B) Domain: $-1 \leq x \leq 1$ and Range: $-\pi \leq y \leq \pi$.

(C) Domain: $-\frac{1}{2} \leq x \leq \frac{1}{2}$ and Range: $-\pi \leq y \leq \pi$.

(D) Domain: $-1 \leq x \leq 1$ and Range: $0 \leq y \leq \frac{\pi}{3}$.

5 What is the value of $\int_0^{\sqrt{3}} \frac{dx}{\sqrt{4-x^2}}$?

(A) $\frac{\pi}{6}$

(B) $\frac{\pi}{3}$

(C) $\frac{\pi}{2}$

(D) $\frac{2\pi}{3}$

6 The expression $\sin 4x + \sqrt{3} \cos 4x$ can be written in the form $2 \sin(4x + \alpha)$.
What is the value of α ?

(A) $\frac{\pi}{6}$

(B) $\frac{\pi}{4}$

(C) $\frac{\pi}{3}$

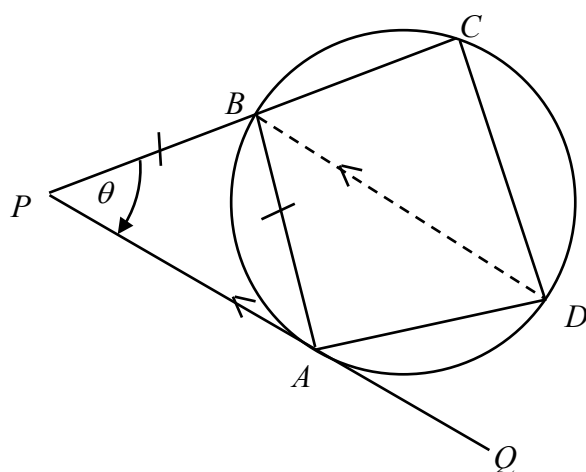
(D) $\frac{\pi}{2}$

- 7 A particle is moving in simple harmonic motion with displacement x .
The acceleration a of the particle is given by

$$a = 25 - 5x.$$

What is the period of motion?

- (A) $\sqrt{5}$
 (B) 5
 (C) $\frac{2\pi}{\sqrt{5}}$
 (D) $\frac{2\pi}{5}$
- 8 In the diagram $ABCD$ is a cyclic quadrilateral. The tangent PQ touches the circle at A . The diagonal BD is parallel to the tangent PQ . QA produced intersects with CB produced at P . $BP = BA$ and $\angle BPA = \theta$.



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What is the size of $\angle BCD$?

- (A) θ
 (B) 2θ
 (C) $180^\circ - \theta$
 (D) $180^\circ - 2\theta$

9 Which of the following is the general solution of $2 \cos 2x = 1$?

(A) $2n\pi \pm \frac{\pi}{6}$

(B) $2n\pi \pm \frac{\pi}{3}$

(C) $n\pi \pm \frac{\pi}{6}$

(D) $n\pi \pm \frac{\pi}{3}$

10 Given that the roots of the cubic equation $4x^3 - 3x^2 - 5x + 2 = 0$ are α , β and γ , what is the value of $\alpha^2 + \beta^2 + \gamma^2$?

(A) $\frac{1}{4}$

(B) $\frac{1}{2}$

(C) $\frac{9}{16}$

(D) $\frac{49}{16}$

End of Section I

Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

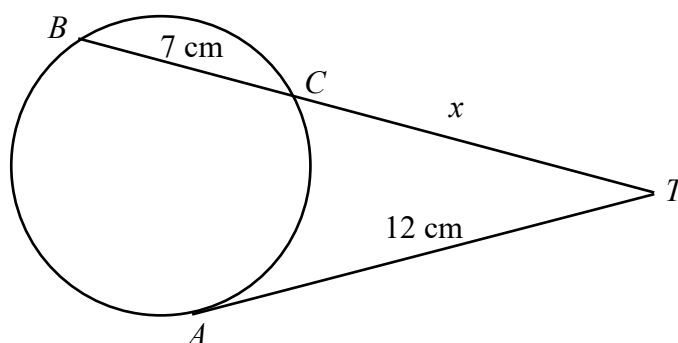
In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet

(a) Solve the inequality $\frac{x}{x-4} \geq 2$. 2

- (b) The point P divides the interval AB externally in the ratio $3 : 1$.
Given the points $A(-3,1)$ and $B(1, -2)$, find the coordinates of P . 2

- (c) In the diagram below, AT is the tangent to the circle at A .
 BT is a secant meeting the circle at B and C . 2



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Given that $AT = 12$ cm, $BC = 7$ cm and $CT = x$ cm, find the value of x .
No reasons are required.

(d) Use the substitution $x = u - 2$ to evaluate $\int_{-1}^2 \frac{3x+5}{\sqrt{x+2}} dx$. 3

Question 11 continues on the following page

Question 11 (continued)

- (e) The function $f(x) = \log_e x - \sin x + 1$ has a zero near $x = 0.75$. **3**

Use one application of Newton's Method to obtain another approximation to this zero.

Give your answer correct to 2 significant figures.

- (f) Find the term independent of x in the expansion of $\left(x^2 - \frac{2}{x}\right)^{12}$. **3**

End of Question 11

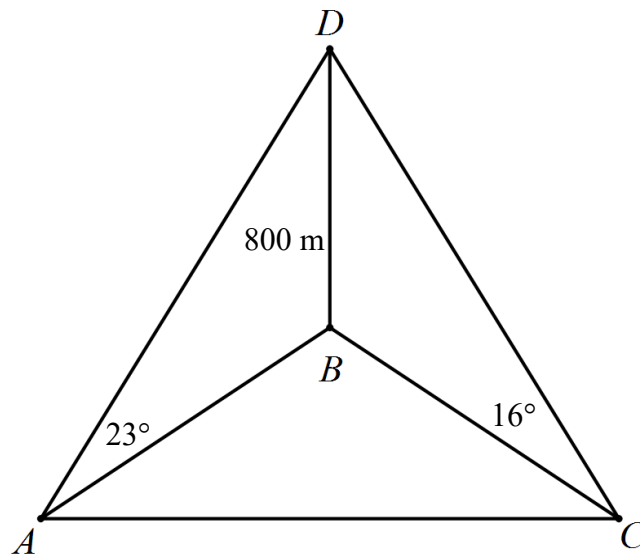
Question 12 (15 marks) Use a SEPARATE writing booklet

- (a) The polynomial $P(x) = 2x^3 + x^2 + ax + 6$ has a zero at $x = 2$.
- (i) Determine the value of a . 1
 - (ii) Find the linear factors of $P(x)$. 2
 - (iii) Hence, or otherwise, solve $P(x) \geq 0$ 1
- (b) Prove by Mathematical Induction that
- $$2 \times 1! + 5 \times 2! + 10 \times 3! + \dots + (n^2 + 1)n! = n \times (n+1)! \text{ for } n \geq 1. \quad 3$$
- (c) Consider the function $f(x) = 6x - 2x^3$.
- (i) Find the largest domain containing the origin for which $f(x)$ has an inverse function $f^{-1}(x)$. 2
 - (ii) Find the gradient of the inverse function at $x = 0$. 2

Question 12 continues on the following page

Question 12 (continued)

- (d) Chris and Aaron are competing in a sailing boat race. Chris (C) can see the top of a vertical cliff (D) that is 800 m above sea level. The cliff is on a bearing of 329° from his position and the angle of elevation to the top of the cliff (D) is 16° . Aaron (A) can also see the top of the cliff on a bearing of 049° with an angle of elevation of 23° . The base of the cliff (B) is at sea level.



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- (i) Show that $\angle ABC = 80^\circ$. 1
- (ii) Find the distance AC between the two sailing boats to the nearest metre. 3

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet

- (a) Consider the point $P(2p, p^2)$ that lies on the parabola $x^2 = 4y$.

- (i) Show that the equation of the normal at P is given by 2

$$x + py - 2p - p^3 = 0.$$

- (ii) The normal meets the y -axis at Q . 2
Find the coordinates of the midpoint M of PQ .

- (iii) Find the locus of the point M . 1

- (b) The acceleration of a particle moving in a straight line is given by

$$a = 2x^3 + 2x$$

where x is the displacement of the particle from the origin at time t seconds.

Initially the particle is at the origin moving at 1 m/s.

- (i) Show that the velocity of the particle is given by $v = x^2 + 1$. 2

- (ii) Hence, or otherwise, find the displacement of the particle 3
after $\frac{\pi}{4}$ seconds.

- (c) Evaluate $\int_0^{\frac{\pi}{2}} \cos^2 x \, dx$. 2

- (d) A rectangle is expanding in such a way that at all times, its length is twice as 3
long as its width. If its area is increasing at a rate of $18 \text{ cm}^2/\text{s}$, find the rate at
which its perimeter is increasing when the width of the rectangle is 80 centimetres.

Question 14 (15 marks) Use a SEPARATE writing booklet

- (a) A frozen cake is removed from a freezer at -10°C and is placed in a room at a constant temperature of 20°C .

Thereafter its temperature T° is changing so that after t minutes

$$\frac{dT}{dt} = K(20 - T) \text{ where } K \text{ is a constant.}$$

- (i) Show that $T = 20 - Be^{-Kt}$ satisfies this condition. 1
- (ii) Find the value of B . 1
- (iii) If, initially, the temperature was increasing at the rate of 3°C per minute, find the value of K . 2
- (iv) Find the temperature of the cake 5 minutes after it was placed in the room. 2
Give your answer to the nearest degree.
- (b) A football is kicked at an angle of α to the horizontal. The position of the ball at time t seconds is given by

$$x = vt \cos \alpha \quad \text{and} \\ y = vt \sin \alpha - \frac{1}{2}gt^2$$

(DO **NOT** PROVE THESE)

where $g \text{ m/s}^2$ is the acceleration due to gravity and $v \text{ m/s}$ is the initial velocity of the football.

- (i) Show that the equation of the path of the football is 2
- $$y = x \tan \alpha - \frac{gx^2}{2v^2} \sec^2 \alpha.$$
- (ii) If $g = 10 \text{ m/s}^2$, $v=20 \text{ m/s}$ and the ball just clears the head of a 1.8 metre tall player that is 10 metre away, calculate the angle(s) to the horizontal at which the football is initially kicked. 3
Give your answer correct to the nearest minute.
- (c) By considering the expansion of $(1+x)^n$ and the value of $\int_0^3 (1+x)^n dx$, 4
show that $\sum_{k=0}^n \frac{1}{k+1} \binom{n}{k} 3^{k+1} = \frac{1}{n+1} (4^{n+1} - 1).$

END OF PAPER

2017 EXT 1 TRIAL HSC

1. $x = 2 + 1$
 $y = + - 2$

$$\therefore + = y + 2$$

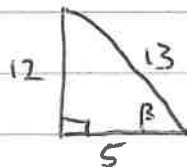
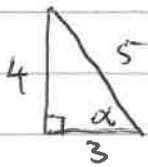
$$x = 2(y + 2) + 1.$$

$$x = 2y + 4 + 1.$$

$$x - 2y - 5 = 0$$

(D)

2.



$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$= \frac{3}{5} \times \frac{5}{13} + \frac{4}{5} \times \frac{12}{13}$$

$$= \frac{63}{65}$$

(D)

3. $m_1 = \frac{2}{3}$ $m_2 = -\frac{1}{2}$

$$\therefore \tan \theta = \left| \frac{\frac{2}{3} + \frac{1}{2}}{1 + \left(\frac{2}{3}\right)\left(-\frac{1}{2}\right)} \right|$$

$$\therefore \theta = 60^\circ$$

(D)

$$4. \quad \frac{y}{3} = \cos^{-1} 2x$$

$$-1 \leq 2x \leq 1 \rightarrow -\frac{1}{2} \leq x \leq \frac{1}{2}$$

$$0 \leq \frac{y}{3} \leq \pi \rightarrow 0 \leq y \leq 3\pi$$

(A)

$$5. \quad \int_0^{\sqrt{3}} \frac{dx}{\sqrt{4-x^2}} = \left[\sin^{-1} \left(\frac{x}{2} \right) \right]_0^{\sqrt{3}}$$

$$= \frac{\pi}{3} \quad (B)$$

$$6. \quad 2 \sin(4x + \alpha)$$

$$= 2 \sin 4x \cos \alpha + 2 \sin \alpha \cos 4x$$

$$\therefore 2 \cos \alpha = 1$$

$$\cos \alpha = \frac{1}{2}$$

$$2 \sin \alpha = \sqrt{3}$$

$$\sin \alpha = \frac{\sqrt{3}}{2}$$

$$\therefore \alpha = \frac{\pi}{3} \quad (C)$$

$$7. \quad a = -5(x-5)$$

$$\therefore -r^2 = -5$$

$$r = \sqrt{5}$$

$$\therefore P = \frac{2\pi}{\sqrt{5}} \quad (C)$$

8. (B)

9. $2 \cos 2x = 1$

$$\cos 2x = \frac{1}{2}$$

$$\therefore 2x = 2\pi n \pm \frac{\pi}{3}$$

$$\therefore x = \pi n \pm \frac{\pi}{6} \quad (C)$$

10. $4x^3 - 3x^2 - 5x + 2 = 0$

$$(\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$$

$$= \left(\frac{3}{4}\right)^2 - 2\left(-\frac{5}{4}\right)$$

$$= \frac{49}{16}$$

(D)

11. a) $\frac{x}{x-4} \geq 2$

Critical points

$$\boxed{x \neq 4}$$

$$\frac{x}{x-4} = 2$$

$$x = 2x - 8$$

$$x = 8$$



test $x = 0$

$$\frac{0}{-4} \geq 2$$

FALSE

$$\therefore 4 < x \leq 8$$

b) $3: -1$

$$x = \frac{-1 \times -3 + 3 \times 1}{2}$$

$$= 3$$

$$y = \frac{-1 \times 1 + 3 \times -2}{2}$$

$$= -3\frac{1}{2}$$

$$P(3, -3\frac{1}{2})$$

$$c) \quad 12^2 = x(x+7)$$

$$144 = x^2 + 7x$$

$$0 = x^2 + 7x - 144$$

$$0 = (x+16)(x-9)$$

$$\therefore x = 9 \quad (x > 0)$$

$$d) \quad x = u - 2$$

$$\therefore \frac{dx}{du} = 1$$

$$dx = du$$

$$u = x + 2$$

$$x = 2 \rightarrow u = 4$$

$$x = -1 \rightarrow u = 1$$

$$\begin{aligned} 3x + 5 &= 3(u-2) + 5 \\ &= 3u - 6 + 5 \\ &= 3u - 1 \end{aligned}$$

$$\therefore \int_1^4 (3u - 1) u^{-\frac{1}{2}} du$$

$$= \int_1^4 3u^{\frac{1}{2}} - u^{-\frac{1}{2}} du$$

$$= \left[2u^{\frac{3}{2}} - 2u^{\frac{1}{2}} \right]_1^4$$

$$= (16 - 4) - (2 - 2)$$

$$= 12$$

$$e) f'(x) = \frac{1}{x} - \cos x$$

$$\therefore x_2 = 0.75 - \frac{\log_e(0.75) - \sin(0.75) + 1}{\frac{1}{0.75} - \cos(0.75)}$$

$$= 0.699 \dots$$

$$= 0.70$$

$$f) \text{ Constant} = \binom{12}{4} (x^2)^4 \left(-\frac{2}{x}\right)^8$$

$$= 126720$$

$$12. (i) 0 = 2(2)^3 + 2^2 + 2a + 6$$

$$0 = 16 + 4 + 2a + 6$$

$$0 = 26 + 2a$$

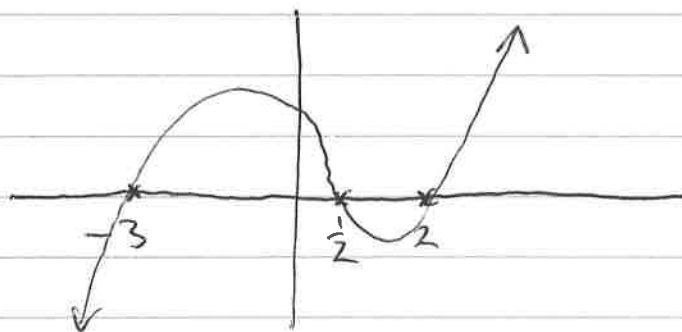
$$\therefore a = -13$$

$$(ii) f(x) = 2x^3 + x^2 - 13x + 6$$

$$\begin{array}{r} 2x^2 + 5x - 3 \\ x-2 \overline{) 2x^3 + x^2 - 13x + 6} \\ \underline{2x^3 - 4x^2} \\ 5x^2 - 13x + 6 \\ \underline{5x^2 - 10x} \\ -3x + 6 \\ \underline{-3x + 6} \\ 0 \end{array}$$

$$\therefore f(x) = (x-2)(x+3)(2x-1)$$

(iii)



$$f(x) \geq 0 \quad -3 \leq x \leq \frac{1}{2}, \quad x \geq 2.$$

b) Prove true for $n=1$.

$$\text{LHS} = 2 \times 1!$$

$$= 2$$

$$\text{RHS} = 1 \times (1+1)!$$

$$= 2$$

$\therefore$ true for $n=1$.

Assume true for $n=k$.

$$2 \times 1! + 5 \times 2! + \dots + (k^2 + 1)k! = k \times (k+1)!$$

If true for $n=k$, prove true for $n=k+1$.

RTP

$$\begin{aligned} 2 \times 1! + 5 \times 2! + \dots + (k^2 + 1)k! + ((k+1)^2 + 1)(k+1)! \\ = (k+1)(k+2)! \end{aligned}$$

$$\text{LHS} = k \times (k+1)! + (k^2 + 2k + 2)(k+1)!$$

$$= (k+1)! \left[k + k^2 + 2k + 2 \right]$$

$$= (k+1)! (k^2 + 3k + 2)$$

$$= (k+1)! (k+2)(k+1)$$

$$= (k+2)! (k+1)$$

$$= \text{RHS}$$

$\therefore$ true for $n=k+1$.

$$c) f(x) = 6x - 2x^3$$

$$f'(x) = 6 - 6x^2$$

$$\text{let } f'(x) = 0$$

$$6 - 6x^2 = 0$$

$$\therefore x = \pm 1$$

$$f''(x) = -12x$$

$$f''(1) = -12$$

$(1, 4)$ is a
max t.p.

$$f''(-1) = 12$$

$(-1, -4)$ is a
min t.p.

$$\text{Domain: } \{x: -1 \leq x \leq 1\}$$

$$(ii) y = 6x - 2x^3$$

$$\frac{dy}{dx} = 6 - 6x^2$$

$$\therefore \text{At } x = 0$$

$$\frac{dy}{dx} = 6$$

$$\frac{dx}{dy} = \frac{1}{6}$$

$$\therefore m = \frac{1}{6}$$

$$d)(i) \angle ABC = (360^\circ - 329^\circ) + 49^\circ$$

$$= 80^\circ$$

$$(ii) \tan 16^\circ = \frac{800}{BC}$$

$$\therefore BC = \frac{800}{\tan 16^\circ}$$

$$\tan 23^\circ = \frac{800}{AB}$$

$$AB = \frac{800}{\tan 23^\circ}$$

$$(AC)^2 = \left(\frac{800}{\tan 16^\circ} \right)^2 + \left(\frac{800}{\tan 23^\circ} \right)^2$$

$$- 2 \left(\frac{800}{\tan 16^\circ} \right) \left(\frac{800}{\tan 23^\circ} \right) \cos 80^\circ$$

$$= 9509760$$

$$\therefore AC = 3083.79$$

$$= 3084 \text{ m}$$

$$13. \quad a) \quad y = \frac{x^2}{4} \quad y' = \frac{x}{2}$$

(i)

$$m_T = \frac{2p}{2}$$

$$= p$$

$$m_N = -\frac{1}{p}$$

$$y - p^2 = -\frac{1}{p} (x - 2p)$$

$$py - p^3 = -x + 2p$$

$$x + py - 2p - p^3 = 0$$

(ii) let $x = 0$

$$py - 2p - p^3 = 0$$

$$py = 2p + p^3$$

$$y = 2 + p^2$$

$$Q(0, 2 + p^2)$$

$$M \rightarrow \frac{0 + 2p}{2}, \quad \frac{p^2 + p^2 + 2}{2}$$

$$(p, p^2 + 1)$$

(iii) $\therefore y = x^2 + 1$

$$b) \quad a = 2x^3 + 2x$$

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 2x^3 + 2x$$

$$\frac{1}{2} v^2 = \frac{1}{2} x^4 + x^2 + C$$

$$\text{when } x=0 \quad v=1$$

$$\frac{1}{2} = C$$

$$\frac{1}{2} v^2 = \frac{1}{2} x^4 + x^2 + \frac{1}{2}$$

$$v^2 = x^4 + 2x^2 + 1$$

$$= (x^2 + 1)^2$$

$$v = \pm (x^2 + 1)$$

$$\text{but when } x=0 \quad v=1$$

$$\therefore v = x^2 + 1$$

$$(ii) \quad \frac{dx}{dt} = x^2 + 1$$

$$\frac{dt}{dx} = \frac{1}{x^2 + 1}$$

$$t = \tan^{-1}(x) + C$$

$$x=0, \quad t=0 \quad \therefore C=0$$

$$\therefore x = \tan(t)$$

$$x = \tan\left(\frac{\pi}{4}\right)$$

$$= 1 \text{ m.}$$

$$c) \int_0^{\frac{\pi}{2}} \cos^2 x \, dx$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{2}} 1 + \cos 2x \, dx$$

$$= \frac{1}{2} \left[x + \frac{1}{2} \sin 2x \right]_0^{\frac{\pi}{2}}$$

$$= \frac{1}{2} \left[\left(\frac{\pi}{2} + 0 \right) - (0 + 0) \right]$$

$$= \frac{\pi}{4}$$

$$d) \text{ width} = x \quad \text{length} = 2x$$

$$A = 2x^2$$

$$\frac{dA}{dx} = 4x$$

$$\frac{dA}{dt} = 18$$

$$\frac{dA}{dt} = \frac{dA}{dx} \times \frac{dx}{dt}$$

$$18 = 4x \times \frac{dx}{dt}$$

$$x = 90$$

$$\frac{dx}{dt} = \frac{18}{320}$$

$$= \frac{9}{160}$$

$$P = 6x$$

$$\frac{dP}{dx} = 6$$

$$\frac{dP}{dt} = \frac{dx}{dt} \times \frac{dP}{dx}$$

$$= \frac{9}{160} \times 6$$

$$= \frac{27}{80} \text{ cm/s}$$

$$14. a) T = 20 - Be^{-kt}$$

$$\begin{aligned}\frac{dT}{dt} &= -Be^{-kt} \times -k \\ &= kBe^{-kt}\end{aligned}$$

$$T = 20 - Be^{-kt}$$

$$Be^{-kt} = 20 - T$$

$$\therefore \frac{dT}{dt} = k(20 - T)$$

$$(ii) t = 0, T = -10$$

$$-10 = 20 - Be^0$$

$$-30 = -B$$

$$\therefore B = 30$$

$$(iii) 3 = k(20 + 10)$$

$$\therefore k = \frac{3}{30}$$

$$3 = k(20 + 10)$$

$$k = \frac{1}{10}$$

$$\begin{aligned}
 \text{(iv)} \quad T &= 20 - 30 e^{-\frac{1}{10} \times 5} \\
 &= 1.8 \dots \\
 &= 2^\circ.
 \end{aligned}$$

$$b)(i) \quad x = v + \cos \alpha$$

$$\therefore t = \frac{x}{v \cos \alpha}.$$

$$y = v \left[\frac{x}{v \cos \alpha} \right] \sin \alpha - \frac{1}{2} g \left[\frac{x}{v \cos \alpha} \right]^2$$

$$= x \tan \alpha - \frac{g x^2}{2 v^2} \sec^2 \alpha$$

$$\begin{aligned}
 \text{(ii)} \quad 1.8 &= 10 \tan \alpha - \frac{10 (10)^2}{2 (20)^2} \sec^2 \alpha
 \end{aligned}$$

$$1.8 = 10 \tan \alpha - \frac{1000}{800} (1 + \tan^2 \alpha)$$

$$1.8 = 10 \tan \alpha - 1.25 - 1.25 \tan^2 \alpha$$

$$1.25 \tan^2 \alpha - 10 \tan \alpha + 3.05 = 0$$

$$\tan \alpha = \frac{10 \pm \sqrt{100 - 4(1.25)(3.05)}}{2.5}$$

=

$$\therefore \tan \alpha = 7.682390528$$

$$\alpha = 82.58 \dots$$

$$= 83^\circ$$

or

$$\tan \alpha = 0.3176094721$$

$$\alpha = 17.62 \dots$$

$$= 18^\circ$$

$$c) (1+x)^n = \binom{n}{0}x^0 + \binom{n}{1}x^1 + \dots + \binom{n}{n}x^n$$

Ans

$$\int_0^3 (1+x)^n dx = \left[\frac{(1+x)^{n+1}}{n+1} \right]_0^3$$

$$= \left(\frac{4^{n+1}}{n+1} \right) - \left(\frac{1^{n+1}}{n+1} \right)$$

$$= \frac{1}{n+1} (4^{n+1} - 1)$$

$$\int_0^3 \left(\binom{n}{0}x^0 + \binom{n}{1}x^1 + \dots + \binom{n}{n}x^n \right) dx$$

$$= \left[\binom{n}{0}x^1 + \binom{n}{1}\frac{x^2}{2} + \dots + \binom{n}{n}\frac{x^{n+1}}{n+1} \right]_0^3$$

$$= \binom{n}{0}3^1 + \binom{n}{1}\frac{3^2}{2} + \dots + \binom{n}{n}\frac{3^{n+1}}{n+1}$$

$$= \sum_{k=0}^n \binom{n}{k} \frac{3^{k+1}}{k+1}$$

$$= \sum_{k=0}^n \binom{n}{k} \frac{1}{k+1} 3^{k+1}$$

$$\therefore \sum_{k=0}^n \frac{1}{k+1} \binom{n}{k} 3^{k+1} = \frac{1}{n+1} (4^{n+1} - 1)$$